

	<i>but there was confusion as to whether the work would be done on the gas or by the gas.</i>
8 (a) (i)	Simple harmonic motion is an oscillatory motion where the acceleration of the object is directly proportional to its displacement from the equilibrium position and directed towards the equilibrium position.
(ii) 1.	The acceleration is always in opposite direction to the displacement. This indicates the presence of a restoring force that pulls the object back to the equilibrium position when it is on either side of the equilibrium position.
2.	The line is not straight hence the acceleration is not directly proportional to the displacement which is a condition for simple harmonic motion.
(b) (i) 1.	A longitudinal wave is one in which the molecules of the wave oscillate along the direction of travel of the wave.
2.	The speed of the wave is the distance moved by the waveform per unit time in the direction of travel of the wave.

	<i>Note: Candidates found this question particularly challenging. Many defined speed as the product of frequency and wavelength.</i>
(ii) 1.	$T = \frac{1}{f} = \frac{1}{835} = 1.2 \times 10^{-3} \text{ s}$
2.	The time interval is half the period = $6.0 \times 10^{-4} \text{ s}$ <i>Note: Many candidates gave the time interval as the period.</i>
3.	$v_{\max} = \omega x_0$ $= 2\pi f x_0$ $= 2\pi (835)(610 \times 10^{-9})$ $= 3.2 \times 10^{-3} \text{ m s}^{-1}$
4.	$E_{\max} = \frac{1}{2} m v_{\max}^2$ $= \frac{1}{2} (5.3 \times 10^{-26}) (3.2 \times 10^{-3})^2$ $= 2.7 \times 10^{-31} \text{ J}$
(iii)	The speed of travel of sound as a wave is much larger than the speed of vibration of the particles in the wave.
(c)(i)	The source of the energy of the sound wave can be from the vibrational k.e. of a loudspeaker cone, if from a drum, the vibrations of the skin of the drum or if from a string in tension, the vibration of the string.
(c)(ii)	The restoring force acting on a gas molecule is due to the pressure difference between where the molecule is with respect to the atmospheric pressure. If the molecule is displaced from its equilibrium point such that there is higher pressure than atmospheric, the pressure difference resulted in a force that will try to restore it to its equilibrium point.